DSA4262 Group Project:

The Singapore Mental Health Tech Challenge

Project Oreka

Supporting Teachers in Addressing School Bullying with an NLP-based Model and Dashboard

Group 6

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GitHub Repository

https://github.com/chrstntan/DSA4262-Group-Project

1. The "Problem Canvas"

Mental health among youth has become an increasingly important public health concern globally, and Singapore is no exception. Adolescence is a critical developmental stage where social environments, particularly schools, play a major role in shaping emotional well-being. Unfortunately, in Singapore, bullying remains a prevalent issue among students. From 2021 to 2025, schools in Singapore recorded an average of 3 bullying cases per 1,000 primary school students and 8 per 1,000 secondary school students (CNA, 2026). Moreover, these statistics capture only reported and cases and are likely to underestimate the true extent of bullying, as many students do not report incidents due to fear, stigma, or a lack of confidence in intervention.

Because students spend a significant portion of their time in school, negative experiences such as bullying can have sustained and far-reaching impacts on their mental health. PBBs (Person Being Bullied) may feel that there is no safe space to escape from the behaviour, which can lead to chronic stress, anxiety, and feelings of isolation. Over time, repeated exposure to bullying has been linked to more severe mental health outcomes such as depression, reduced self-esteem, and even suicidal ideation. Teenagers who were bullied are nearly twice as likely to experience symptoms of anxiety or depression when compared with teenagers who were not bullied (Haile et al., 2024). As such, examining school bullying through a data-driven mental health lens is crucial, especially in Singapore, where bullying is not uncommon.

Although students are the ones directly affected by bullying, teachers are often the primary adults responsible for responding to students' wellbeing concerns within schools, as they are uniquely positioned at the intersection of students' daily experiences and institutional support systems. Since they interact most frequently with students, teachers are often the first to observe changes in behaviour, social dynamics, or emotional distress. This places them in a critical role not only in identifying but also in handling bullying incidents and mitigating their psychological impact.

1.1 Problem Statement

However, effectively responding to bullying is not always straightforward. Teachers must interpret complex social situations, assess the severity of incidents, and decide when different levels of intervention or escalation are necessary. This challenge is compounded by the realities of their workload. Teachers in Singapore work an average of 47 hours per week (OECD, 2025) (among the highest globally), yet less than one-third of this time is spent on actual classroom teaching, with the remainder devoted to administrative duties, lesson preparation, and student management. As teachers are expected to take on multiple roles beyond academic instruction, including counselling students, managing classroom behaviour, and communicating with parents, the complexity of their responsibilities increases significantly (Tan, 2025). Unsurprisingly, heavy workload has been identified as a key source of stress, with more than 1 in 4 teachers reporting high levels of work-related stress, exceeding that of their international peers. These constraints have direct implications for how effectively teachers can respond to bullying incidents. Research on student-perceived

teacher intervention effectiveness shows that consistent and timely responses by teachers were associated to reductions in bullying, while intermittent or absent responses often allow situations to persist or worsen instead (Muñoz-Fernández et al., 2026). Even when teachers are aware of bullying cases, competing demands on their time and attention may hinder their ability to intervene in a consistent and sustained manner.

Our problem statement:

Teachers often struggle to intervene early in bullying incidents because they must manage numerous competing responsibilities, allowing harmful behaviours and wellbeing risks to escalate before timely support can be provided.

As mentioned by Jason Tan, an associate professor at the National Institute of Education, teachers must juggle and interact with hundreds of students across form class, subject teaching and co-curricular activities, and at the same time attempt to tell intentional bullying apart from ordinary rudeness or thoughtlessness that may not constitute a desire to inflict intentional harm. This makes it especially difficult to gather sufficient information and respond appropriately under heavy workloads and tight time constraints.

1.2 Jobs To Be Done

With these concerns of teachers in mind, we propose the following JTBD:

When multiple bullying cases arise at the same time, I want to prioritise which cases require urgent attention so that I can allocate my time and resources effectively, without overlooking students whose situations require more urgent interventions.

In practice, teachers are often faced with multiple competing cases. Without a clear and structured way to assess these cases, they must rely on subjective judgment under time pressure, increasing the risk of inconsistent or delayed intervention. As a result, critical cases may not receive timely attention, allowing harm to escalate, while less urgent cases may consume disproportionate time and resources. This creates not only inefficiencies in workload management, but also potential risks to student wellbeing. Therefore, there is a need for a systematic framework that supports teachers in making informed, consistent, and timely prioritization decisions, enabling earlier intervention for high-risk cases while maintaining fair and balanced attention across all reported incidents.

2. The Framework and Model Prototype

To support timely and effective intervention in school bullying cases, we propose the following 4-step framework that transforms bullying reports into actionable signals for decision-making:

1. PBB/witness submits reports of bullying experienced/witnessed using our reporting form
2. Obtain out signals (Frequency, Severity and Impact) from the reports and model
3. Assign an urgency level to each report to determine the recommended actions
4. Display cases to handle in teacher's dashboard by urgency level (and send alerts for severe cases, if present)

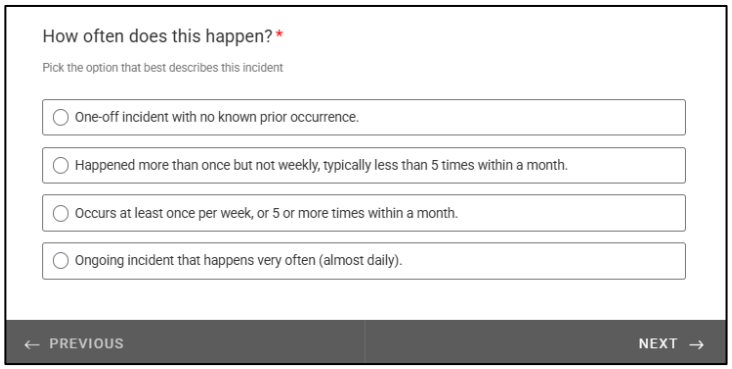
The system is designed to lower reporting barriers, especially for bystanders or PBBs who may fear retaliation, while still enabling teachers to identify and prioritise cases that require attention.

2.1 Reporting Platform

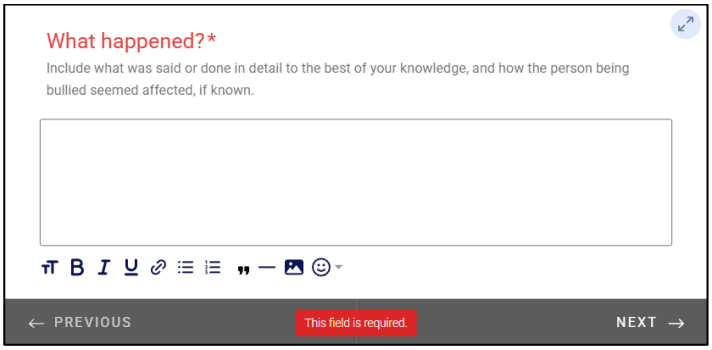
We first introduce a reporting platform where students (both witnesses and PBBs) can submit accounts of bullying incidents they have encountered.

Of the questions asked, the 3 questions in our reporting platform that correspond to our 3 main signals are as follows:

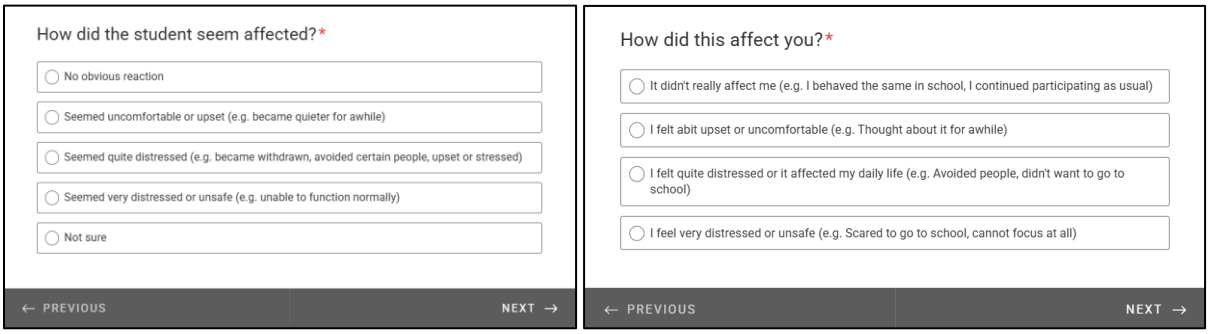
5. To assess frequency of bullying:



6. To assess severity of bullying:



7. To assess impact of bullying:



To access the full reporting form, please visit the following link: Reporting Form.

2.2 Our Signals

From the report submissions, we focus on extracting 3 primary signals: bullying frequency, impact and severity. These signals were inspired by a New Zealand's Bullying Assessment Tool that was created by Bullying-Free NZ, a national initiative backed by the New Zealand Ministry of Education that supports schools in implementing consistent, evidence-based approaches to bullying prevention and response. In our framework, we adopt this assessment tool as a foundational template, particularly its emphasis on structured categorization of bullying behaviours and patterns. However, rather than applying it directly, we adapted the definitions within each category to align more closely with the types of incidents likely to be reported through our student-submitted narratives. Importantly, these adaptations were informed by consultations with local stakeholders, including an educator from MOE and members of the Singapore Children's Society. Their input helped ensure that the framework reflects the realities of school environments in Singapore. Through this process, we grounded our signal design not only in established theory, but also in practical, context-specific expertise.

2.2.1 Frequency

The first signal, bullying frequency, captures whether the incident is isolated or part of a repeated pattern over time. Research consistently shows that persistent exposure to bullying is linked to adverse psychological outcomes, including anxiety, depression, and social withdrawal (Man et al., 2022). Consequently, even incidents of lower severity may warrant intervention if they occur frequently, as they signal sustained harm and increased long-term risk.

Frequency is assessed through a single-selection question in the reporting form, where reporters indicate how often the bullying incident has occurred. Below is the frequency scoring framework:

Frequency	Description	Score

Rare	A one-off incident with no known prior occurrence.	1
Occasional	Reported to have happened more than once but not weekly, typically less than 5 times within a month.	2
Frequent	Occurs at least once per week, or 5 or more times within a month.	3
Persistent	An ongoing incident that happens almost daily.	4

We chose 5 to be the differentiating number between 'Occasional' and 'Frequent' as within a typical four-week school month, 5 incidents likely implies that the behaviour is happening more than once a week on average, which goes beyond occasional conflict and starts to reflect a consistent pattern of targeting.

2.2.2 Impact

The second signal we assess is the impact of the bullying, which captures the extent to which the incident affects the student's emotional, psychological, and daily functioning. This ranges from low impact, where the student may experience temporary upset with no lasting effects, to severe impact, where there is significant distress affecting their safety and daily functioning.

Impact is also assessed through a single-selection question in the reporting form, where PBBs indicate how the bullying affected them (or how the student being bullied seemed affected, in cases where the reporter is a witness and not the PBB). Below is the impact scoring framework:

Impact	Description	Score
Low	Minimal or no observable emotional or behavioural impact.	1
Moderate	Mild emotional discomfort without significant observable disruption to functioning. (e.g. thinking/worrying about it for a while)	2
High	Clear emotional distress with noticeable behavioural changes. (e.g. avoidant behaviour, constant worry)	3

Severe	Severe distress or perceived lack of safety, or extreme behavioural changes, requiring immediate support. (e.g. afraid of going to school, lack of appetite)	4

2.2.3 Severity

The third signal, bullying severity, reflects the intensity of the harmful behaviour described in the report, including indicators such as threats, harassment, or verbal abuse. Higher severity incidents are associated with greater emotional distress, fear, and potential trauma for the victim. As such, they serve as strong indicators for more urgent intervention.

Unlike the previous signals, severity is derived from an open-ended text response provided in the reporting form, where students describe what happened in their own words. Each report will then be processed using a BERT-based natural language processing model, which assigns the corresponding severity score based on the content of the submission. We define a severity scoring framework presented below that outlines the implications of each score level.

Level	Examples	Score
Mild	Teasing, name calling, social exclusion, rumour spreading, taking (and returning) belongings as prank	1
Moderate	Threats, blackmailing, intimidation (non-physical coercion), damaging / taking (and not returning) of belongings, demanding pocket money	2
Major	Serious physical threats and aggressive gestures (raising fists, cornering), minor physical harm (e.g. pushing, shoving, rough handling during “play” that causes harm)	3
Severe	Physical harm requiring medical attention (e.g. causing bleeding, sprains), using objects during conflict (e.g. sharp objects like scissors to threaten or harm), encouraging/engaging in self-harm or suicidal behaviours	4

The mild level (score 1) includes behaviours that, while harmful, are typically lower intensity and may be situational or less sustained. These actions often impact a student’s social experience and self-esteem but do not immediately indicate severe psychological or physical danger. The moderate level (score 2) captures behaviours that introduce elements of coercion and intimidation. These actions

represent a shift from passive harm to more deliberate and targeted aggression. The major level (score 3) involves credible threats and harassment while the severe level (score 4) is reserved for the most critical cases that pose a threat to a student's safety.

2.3 Our Model for Assessing Severity

Since bullying frequency and impact is explicitly self-reported through the structured dropdown field in the reporting form, our model will be focused on assessing only the severity of each report as mentioned above. This design choice reflects the nature of each signal. Frequency and impact are subjective, experiential measures that may not always be fully observable from the textual incident recount provided alone. Having predefined options for students ensures consistency and standardisation across responses, reducing ambiguity and possible lack of mention.

On the other hand, severity depends on case details and how serious the incident is in nature. These distinctions are often expressed in varied ways across different reports, making them difficult to standardise using simple dropdown choices. As a result, using BERT directly on the textual recounts helps ensure that severity is assessed more objectively and consistently.

2.3.1 Data Generation

We developed a labelled dataset tailored to the local student context. As access to actual school bullying reports is limited, we sourced bullying recounts from Singapore-based online community platforms, including relevant discussion threads on Reddit (e.g., r/singapore, r/sgexams) as well as mental health forums such as mindline.sg. These sources provided authentic, experience-based narratives that reflect the language, scenarios, and social dynamics present in local school environments. To improve consistency and suitability for model training, we leveraged large language models (LLMs) to standardize and reframe these recounts into a "report-style" format and generate variations of the same incident from different perspectives (either victim or eyewitness) while preserving the original meaning and context. This allowed us to enrich the dataset with different perspectives, better reflecting the range of inputs expected in a reporting system.

Each report was then manually annotated by all five team members. For every entry, we independently assigned both a frequency and severity score based on the scoring framework for these signals mentioned in the previous section. The final label for each report was determined by the mode class of the group's votes. To prevent class imbalance, we curated it to achieve a relatively balanced distribution across categories, targeting approximately 25 reports for each (frequency, severity) combination (total of 16 combinations since each signal is scored between 1 to 4). This resulted in a total of 446 labelled reports.

2.3.2 Model Training, Validation and Testing

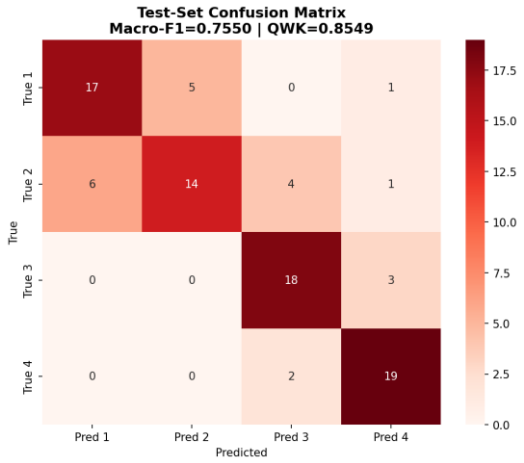
To predict the severity of bullying reports, we chose to use MentalRoBERTa, which was further trained on our dataset of 446 bullying recounts. Instead of traditional machine learning models (e.g., Logistic Regression or SVM), we adopted a transformer-based approach because bullying severity depends heavily on context, tone, and semantics, rather than just the presence of specific keywords. Transformer models capture these nuances better through contextual embeddings, making them more suitable for this task.

MentalRoBERTa is pretrained on mental health–related Reddit text data, including posts from subreddits like *r/offmychest* and *r/mentalhealth* (Ji et al., 2021). Since our project also uses Reddit-based text data, the model's pretraining corpus is closely aligned with our data distribution, allowing it to generalize better and capture relevant patterns more effectively. This is important because bullying reports are often informal, narrative-based, and contain emotional cues and implicit harm. Therefore, MentalRoBERTa is well-equipped to understand bullying reports due to its domain-specific pretraining.

MentalRoBERTa was chosen over MentalBERT as RoBERTa's dynamic masking (which randomly re-selects masked tokens each training epoch rather than fixing them) combined with its larger pretraining corpus, results in stronger contextual representations and better generalisation. This would allow the model to produce better contextual representations, generalize more effectively, and perform better on bullying severity classifications.

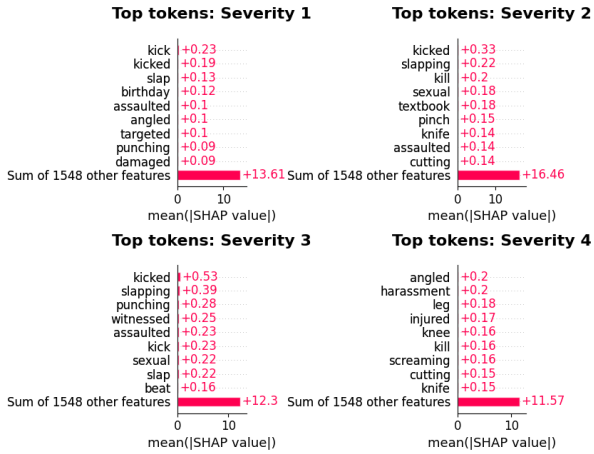
Using a stratified 80/20 train-test split to preserve the class distribution across train test sets, the test set was kept locked until final evaluation to avoid any data leakage. On the training set, 5-fold stratified cross-validation was conducted, re-initialising MentalRoBERTa from scratch each fold. Stratified cross-validation was used to ensure that each training/validation fold maintained the same balanced class distribution as the entire dataset, preventing biased performance evaluation. It also provided a more accurate and stable estimation of model performance on unseen data.

Macro-F1 was used as our main metric due to the dataset having balanced distribution of classes as well as to ensure that all classes were evaluated equally strictly. The cross-validation results were reasonably stable across folds (macro-F1: 0.6955, 0.6642, 0.7644, 0.7298, 0.7212), with a mean of 0.7150 ± 0.0336. Fold 2 was the weakest at 0.6642, likely due to slightly harder validation samples in that split.



Our final model achieved a macro-F1 score of 0.755 on the held-out test set, which was consistent with cross-validation performance and showed no sign of overfitting.

The Quadratic Weighted Kappa, which measures model prediction performance across classes, also shows a highly positive score of 0.855, indicating a high level of agreement between the actual severity labels and the model's predictions. Based on the confusion matrix, misclassifications were largely concentrated between Severity 1 and 2, with five Severity 1 cases predicted as Severity 2 and six Severity 2 cases predicted as Severity 1. This can be attributed to the subjectivity of the severity scoring framework, where the distinction between a 'minor' and 'moderate' incident is less clearly defined compared to higher severities. This confusion matrix also indicates that the model overpredicts (14 cases) more often than it underpredicts (8 cases), which aligns with our objective of prioritizing intervention. In the context of bullying, overpredicting severity is preferable to understating it as this conservative bias ensures that misclassified high-risk cases do not go unnoticed, thereby preventing potential escalation and more grave consequences.

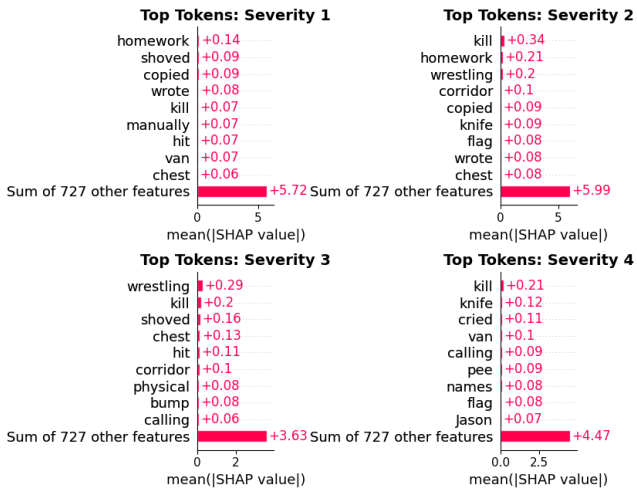


To interpret the model's predictions, SHAP (SHapley Additive exPlanations) was used to identify the most influential tokens for each severity class. The results show that the model relies on semantically meaningful features, with words related to physical aggression such as "kick," "slap,"

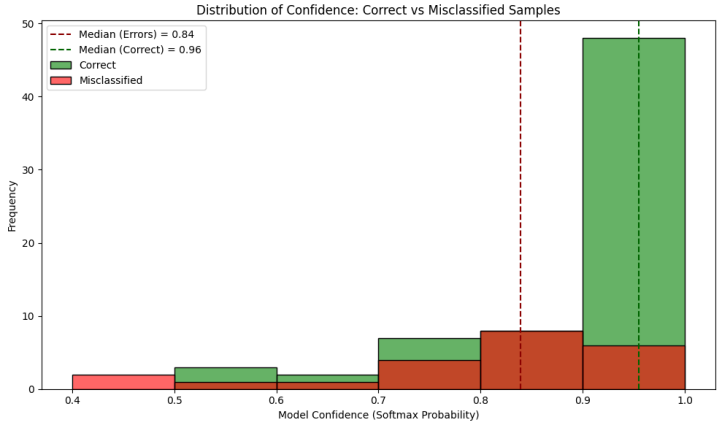
and “assaulted” contributing strongly across classes, while more severe terms like “injured,” “knife,” and “kill” are associated with higher severity levels, indicating that the model has learned a clear progression of severity. However, there was noticeable overlap in important features for mid-level classes (Severity 2 and 3), which explains the model’s relatively lower performance in distinguishing between them. Overall, the presence of many contributing features also reflects the model’s reliance on contextual understanding rather than a small set of keywords.

2.3.3 Error Analysis

2.3.3.1 Global error analysis



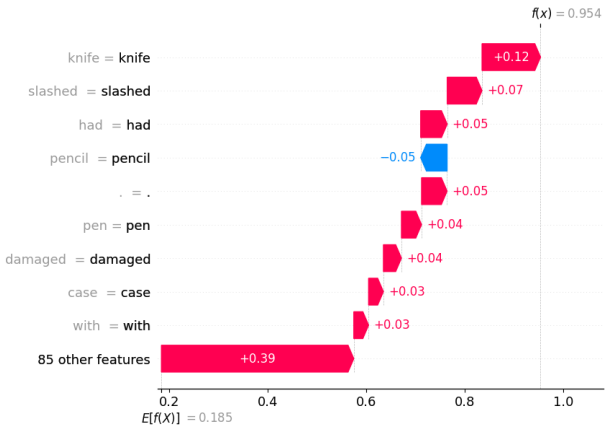
SHAP was applied to the subset of misclassified data points to identify the tokens most influential to the model's decision-making across severity classes. Tokens such as 'kill', 'wrestling', and 'shoved', which indicate violent acts, appeared prominently across multiple classes, suggesting some model ambiguity regarding physical violence. Additionally, the presence of 'homework' is noteworthy, reflecting the specific context of school-related bullying.



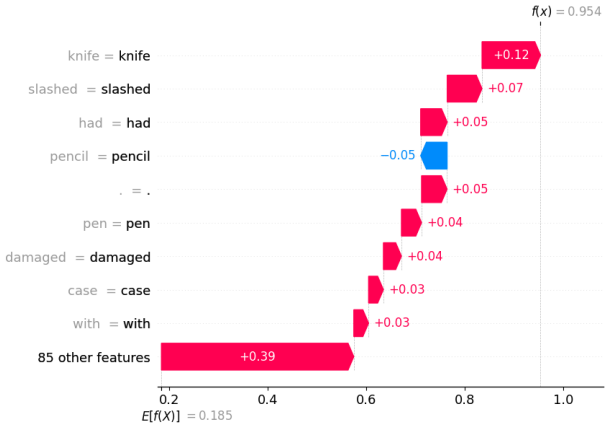
SoftMax probabilities for the test set were aggregated into 'correct' and 'misclassified' subsets. As shown in the distribution plot on the right, correct predictions occur with higher confidence (median = 0.96), whereas errors typically exhibit lower confidence (median = 0.84). Notably, a small number of high-confidence incorrect predictions suggest that the model can be overconfident in ambiguous cases. This reinforces the necessity of a human-in-the-loop approach to verify model outputs.

2.3.3.2 Granular error analysis

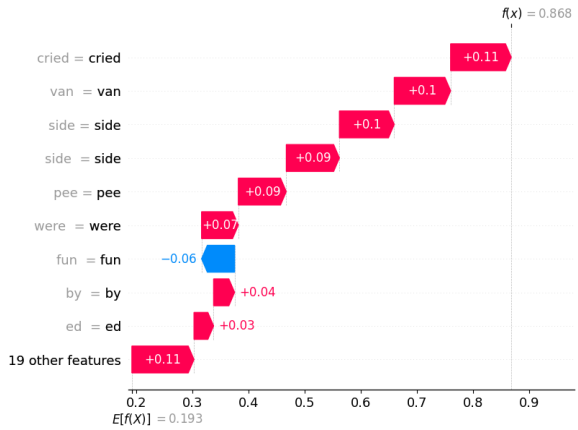
Further SHAP analysis was conducted for datapoints that were predicted by the model to be different by at least 2 levels compared to their assigned severity level.



<u>“Sometimes I will realise that my bag has missing or damaged things. The first time my water bottle was gone and I found it later in the toilet bowl. The second time someone had slashed my pencil case with a penknife. (...) My mum asked why I keep losing things and I didn't know what to say. I feel very unsafe leaving my bag unattended even for a minute.”</u>



This case was manually assigned severity level 2 but predicted as severity level 4 by the model. To diagnose the model's decision-making process at a granular level, we utilized SHAP waterfall plots. These visualizations decompose the prediction for a single instance, starting from the base value, the average model output and illustrate how individual tokens contribute to the final probability. In this specific case, the model predicted Severity Class 4 with high confidence. The tokens ‘knife’ and 'slashed' provided the strongest positive attribution, suggesting the model prioritized threat-based vocabulary to support high severity. While this is in line with our severity framework, it shows the model inadequately understands the context in which the threat-based vocabulary is used, which is not for actual violence on the PBB but rather on their items.



"Evan peed on my shoes for fun when we were peeing side by side. I was so upset that I cried."

This case was manually assigned severity level 1 but predicted as severity level 4 by the model.

This misclassification is primarily attributed to input sparsity. With the text comprising only 28 tokens, the model's self-attention mechanism is forced to distribute weight across a limited feature set. This results in an over-concentration of attribution on high-variance emotive tokens like 'cried'. Without broader contextual qualifiers that might have signalled a lower severity, the model over-indexes on these sparse triggers, erroneously scaling the predicted severity to level 4.

2.4 Signal to Decisions

To recap, each signal is derived from a different source: frequency and impact are explicitly self-reported by students, while severity is extracted from the report text using a BERT-based model. We combine the 3 signals into a single urgency score using a weighted aggregation formula:

$$urgency\ score\ =\ (0.5 \times S)\ +\ (0.25 \times F)\ +\ (0.25 \times I)$$

This enables the system to evaluate each case holistically by integrating multiple dimensions of bullying, rather than considering them in isolation. The weighting scheme is intentionally designed to reflect both the relative reliability and decision-making relevance of each signal in the context of teacher intervention.

Severity is assigned the highest weight as it provides the most direct and objective indication of the seriousness of the incident. It captures the nature of the behaviour itself (e.g., threats, physical harm, coercion), which is a strong predictor of immediate risk and therefore most critical for prioritisation. Frequency is assigned a lower weight as it primarily reflects repetition over time. While repeated incidents may indicate persistence or escalation, high frequency alone does not necessarily correspond to high immediate harm, especially in cases where incidents are relatively more minor in nature. As such, it serves as a supporting rather than dominant indicator of urgency. Impact is also weighted more conservatively to account for variability in reporting sources. Since reports may be

submitted by either PBBs or witnesses, perceived impact can be subjective and influenced by limited or indirect observation. To reduce potential over-reliance on second-hand interpretations, its contribution is calibrated to complement, rather than outweigh, the more objective severity signal.

Once the urgency score is computed, it is mapped to four tiers: low, medium, high, and severe, using the ranges **1.00 – 1.75, 1.75 – 2.50, 2.50 – 3.25, and 3.25 – 4.00**. These cut-offs are intentionally set at regular intervals of 0.75 across the 1–4 scoring scale. We chose evenly spaced thresholds in hopes to improve consistency and transparency in decision-making. Teachers can more easily understand how scores translate into action tiers, and small variations in input signals do not lead to disproportionate jumps between categories.

2.4.1 Low Urgency

Cases classified as low urgency typically involve isolated or lower-severity incidents that do not yet indicate sustained harm but still warrant attention. Our recommended approach for such cases focuses on early, preventative intervention:

8. Teachers should monitor the situation over time to detect any emerging patterns, while reinforcing positive behaviour through clear verbal reminders or class-wide expectations
9. A brief, private check-in with the targeted student can help assess their well-being and ensure they feel supported.

At this stage, the goal is to address the behaviour to de-escalate potential tensions and prevent recurrence without formal disciplinary escalation.

2.4.2 Medium Urgency

Medium-urgency cases reflect either repeated lower-severity incidents or one-off situations with moderate impact, indicating a need for more structured intervention.

10. Teachers are encouraged to conduct direct, individual check-ins with both the target and the aggressor to better understand the context and perspectives involved.
11. Restorative practices play a key role at this level, such as facilitating guided conversations or apologies aimed at repairing harm and rebuilding relationships.
12. Proportionate disciplinary actions such as detention or formal reprimands may be applied to reinforce accountability.
13. Parental involvement is recommended at this stage to ensure alignment between school and home in addressing the behaviour.

The objective is to intervene early and effectively before the situation escalates further.

2.4.3 High Urgency

High-urgency cases involve either severe one-off incidents or repeated patterns of harmful behaviour that pose a significant risk to the student's safety and well-being.

14. Immediate steps must be taken to ensure the safety of the targeted student, including separating involved parties if necessary.
15. Cases should be escalated to the school management team (e.g., discipline master, year head) to ensure coordinated oversight and response.
16. Formal disciplinary procedures may be initiated, including suspension or other serious consequences, depending on the nature of the incident.
17. Immediate documentation at this stage to track actions taken and support any further escalation.
18. Parental involvement is essential to ensure that all stakeholders are informed and engaged.

The focus here is on containment, accountability, and preventing further serious harm.

2.4.4 Severe Urgency

Severe-risk cases represent the most critical situations, often involving extreme forms of bullying such as threats of serious harm, physical violence, or behaviour with potential legal implications.

19. Ensuring the immediate safety of the targeted student is the top priority, which may involve urgent protective measures and removal from unsafe environments.
20. Cases require escalation to senior school leadership, including the principal, and should trigger referrals to school counsellors or external mental health professionals to address potential trauma.
21. Depending on the severity and nature of the incident, teachers and school leaders may also need to consider reporting to external authorities, such as law enforcement, in accordance with school policies and legal obligations.
22. Parents or guardians must be involved immediately to ensure a coordinated and comprehensive response.

At this level, the goal is not only to stop the behaviour but also to provide intensive support and safeguard the long-term well-being of the victim.

2.5 Metrics

To assess the effectiveness of the proposed model prototype, we define three evaluation metrics that collectively capture the timeliness of intervention, teacher experience and reduction of repeat bullying occurrences.

2.5.1 Teacher Response Time

This metric measures the time taken for a teacher to act on a reported case after it has been submitted.

$$Response\ Time\ =\ datetime(Status\ set\ to\ 'in\ progress')\ -\ datetime(Report\ Submitted)$$

It evaluates how effectively the urgency scores guide teachers in prioritising cases. A well-designed urgency scoring system will make high-priority cases more salient, resulting in shorter response times for high-urgency reports compared to lower-urgency ones.

2.4.2 Bi-annual Teacher Survey

A structured survey will be administered twice a year to assess whether the prototype meaningfully reduces teachers' burden and efficiently helps them to handle bullying cases. The survey will measure the following:

Quantitative Responses:

23. Time savings: "How much time does the dashboard save you in managing bullying cases?" (1–10 Likert scale or hours per week)
24. Ease of tracking follow-ups: "How easy is it to monitor the progression of cases over time?" (1–10 Likert scale)
25. Confidence in decision-making: "How confident do you feel in prioritizing interventions based on the dashboard?" (1–10 Likert scale)
26. Usefulness of alerts/notifications: "How effective are the alerts to help you respond to cases promptly?" (1–10 Liker scale)
27. Effectiveness in addressing bullying: "How effective is the dashboard in reducing bullying incidents?" (1–10 Likert scale)
28. Effect on students' mental health: "To what extent has the dashboard helped you identify students at risk and provide timely support?" (1–10 Likert scale)

Qualitative Responses:

29. "What features of the dashboard are most helpful?" (Free response question)
30. "What improvements would make the dashboard more effective in managing bullying cases?" (Free response question)

The teacher survey serves as a practical evaluation metric for the model prototype by capturing its real-world impact on end users. By combining quantitative ratings and qualitative feedback, the survey ensures that the prototype is not only technically sound but also effective, user-friendly, and meaningful in everyday school contexts.

2.4.3 Reduction in Repeat Bullying Cases

This metric determines whether the frequency of bullying cases for the victim has decreased after teacher intervention. A follow-up survey will be sent to the original reporter one month after the initial report, asking whether the bullying has ceased, improved, or persisted.

2.5 Success Criteria

The following success criteria have been defined to evaluate whether the model prototype is functioning as intended:

Dimension	Metric	Target
Urgency Matrix Coverage	Proportion of submitted reports that receive a valid urgency classification without manual override	≥95% of reports
Response Time Improvement	Average teacher response time relative to pre-dashboard baseline	Meaningful reduction in mean response time for high-urgency cases
Survey Sentiment	Proportion of teachers reporting the dashboard reduces prioritisation difficulty	≥70% responding positively

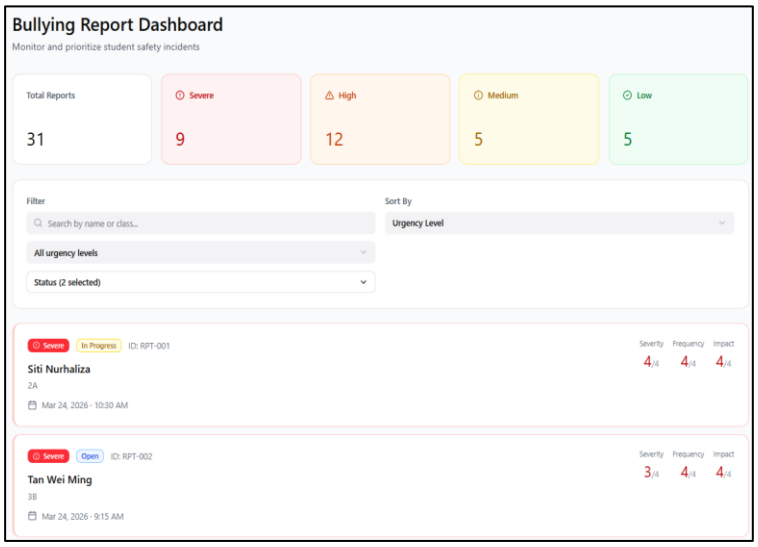
3. The Product Interface

Our product interface is designed to streamline the process of reporting, assessing, and managing bullying cases. It aims to reduce teachers' workload by centralising information, prioritising cases based on urgency, and providing clear guidance for intervention. In addition, the system reduces workload by standardising the process from reporting through to intervention, ensuring consistency in how cases are handled. While structured guidance is provided, teachers retain the flexibility to adjust interventions based on their judgment and the specific context of each case.

3.1 Dashboard

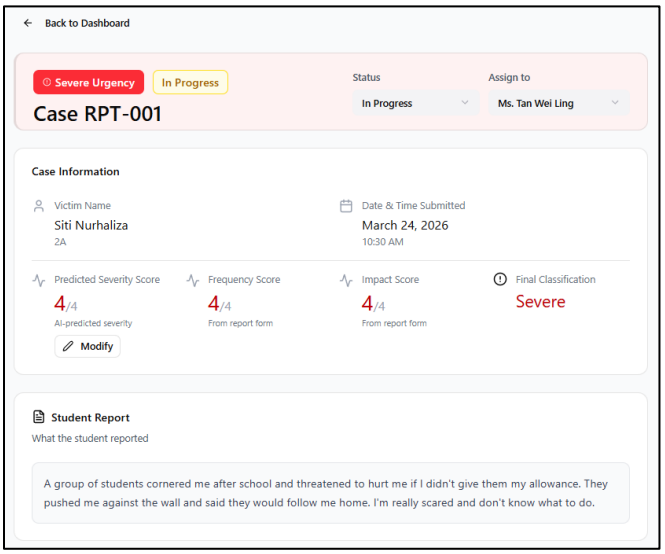
The dashboard acts as the central interface for teachers to manage all reported cases. It consolidates case information into a single view, allowing teachers to quickly understand the overall situation and prioritise their attention based on urgency. By presenting severity levels, risk scores, and workflow status, the dashboard supports efficient decision-making while reducing the need for manual sorting and tracking. Additionally, alerts are triggered for high-risk cases to ensure timely intervention.

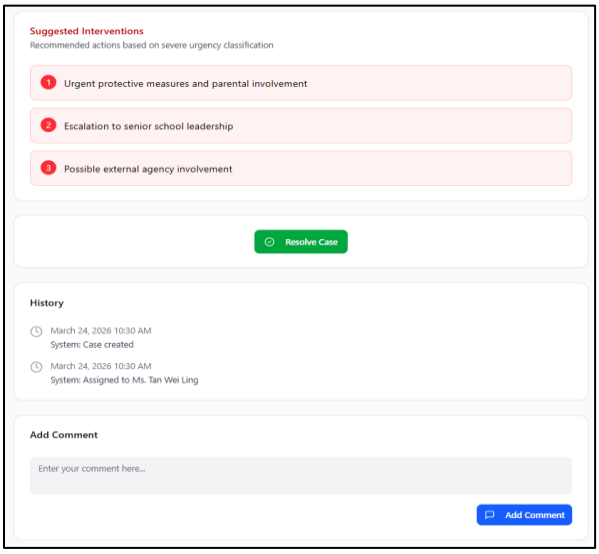
3.1.1 Main Page



The main section of the dashboard provides teachers with an overview of all reported cases. A summary of key case information is displayed at the top, allowing teachers to quickly understand the overall situation. Cases are automatically ranked by urgency, ensuring that higher-risk cases are surfaced first and receive timely attention. This view helps teachers efficiently allocate their time and focus on the most critical situations given their workload constraints.

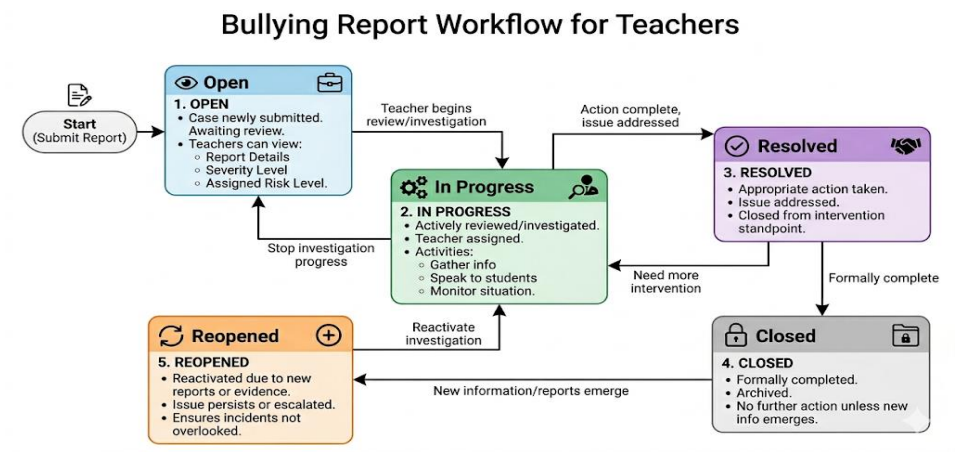
3.1.2 Case Page





When the teacher selects a specific case, they are directed to a detailed case view that presents key information in a structured and accessible format. This includes the status of the case along with the student report, enabling the teacher to quickly understand the context. The system also displays recommended interventions tailored to the assigned urgency level according to our framework, guiding teachers on appropriate next steps. In addition, the case interface includes a history log and comment section for documentation and tracking purposes. Teachers can record actions taken, add notes, and update the case as new information emerges. This creates a centralised record of the case progression, enabling better continuity and accountability over time.

3.1.3 Workflow



To support effective handling of bullying cases, the system also makes use of the following workflow that that tracks each report from initial submission to resolution. This ensures accountability, visibility, and consistent follow-up. After a case is displayed on the dashboard, it enters a workflow pipeline that guides how teachers manage the case.

Workflow Stages

31. **Open**: The case has been newly submitted and is awaiting review. At this stage, teachers can view the report details, severity level, and assigned risk level.
32. **In Progress**: The case has been assigned to a teacher and is actively being reviewed or investigated. This may involve gathering additional information, speaking to involved students, or monitoring the situation.
33. **Resolved**: Appropriate action has been taken, and the issue has been addressed. The case is considered closed from an intervention standpoint.
34. **Closed**: The case is formally completed and archived. No further action is required unless new information emerges.
35. **Reopened**: The case is reactivated if new reports or evidence suggest that the issue persists or has escalated. This ensures that repeated or ongoing incidents are not overlooked.

4. Safety and Ethics Discussion

4.1 Risk of Misclassification

The model may assign an incorrect severity level due to limitations in training data or the inherent ambiguity of free-text reports. When this misclassification is mapped onto the severity scoring framework, it may result in an inaccurate assessment of risk level and potentially leading to interventions that are disproportionate to the actual severity of the incident. Underestimation may

result in serious incidents not being prioritized in a timely manner. Conversely, overestimation may trigger unnecessary escalations, placing undue stress on stakeholders and straining school resources.

To mitigate these risks, the system is designed as a decision-support tool rather than a final decision-maker. Teachers retain full authority to review, interpret, and override model outputs where necessary. This human-in-the-loop approach ensures that professional judgment remains central, while the model serves to assist, rather than replace, the decision-making process.

4.2 Privacy & Confidentiality

Reports may contain sensitive and confidential personal information about students, including identities, relationships, and detailed descriptions of incidents. This creates a risk of unauthorised access, data misuse, or breaches of confidentiality, which could harm students' privacy, damage trust in the reporting system, and discourage future reporting. Improper handling of such data could also expose students to stigma, retaliation, or emotional distress.

Thus, access to sensitive report data should be governed through role-based access control (RBAC), whereby permissions are granted based on specific professional roles and responsibilities. This ensures that only authorised personnel such as teachers, counsellors, or school leaders can view or act on specific cases. Furthermore, schools must establish comprehensive staff training and data-handling protocols to ensure the system is used responsibly and to minimize the risk of accidental data disclosure

4.3 Risk of False Reporting

Students may submit false, exaggerated, or misleading reports, whether intentionally or due to misinterpretation of events. Such reports can unfairly implicate other students, trigger unnecessary or disproportionate interventions, and undermine trust in the system over time. If not carefully managed, this may also divert attention and resources away from genuine cases that require urgent support.

To mitigate this, protocol should require teachers to triangulate information by interviewing involved parties, consulting witnesses, and reviewing historical incident data. This helps build a more complete and accurate understanding of the situation. The system should also require teacher validation before any formal action is taken, ensuring that no intervention is based solely on a single unverified report. The system supports this by providing a case log where teachers can document verification steps and supporting evidence before finalizing an intervention.

4.4 Ethical Considerations

4.4.1 Preserving Human Context

While the automated scores help to prioritise cases, the model may not fully capture the complexity of each situation. To address this, students' written reports are included alongside the model outputs, ensuring that decision-makers can interpret each case within its full qualitative context. This supports more informed and empathetic decision-making and prevents the over-simplification of sensitive situations.

4.4.2 Transparency to Students

Transparency is essential in maintaining trust. Students will be notified that an automated system assists in processing their reports, ensuring transparency regarding how their data is analysed. This promotes informed participation and encourages responsible reporting.

5. Implementation Barriers

5.1 Public Perception and Parent Scrutiny

The use of labels and scores to quantify bullying incidents may be perceived as insensitive by the public, particularly by parents who expect a more nuanced and human-centred approach. In addition, the use of AI in handling student wellbeing may raise concerns about reliability and appropriateness, leading to scepticism about the system. To address this, it is important to clearly communicate that the system is a decision-support tool, with final decisions made by teachers.

5.2 Psychological Barrier

Teachers may perceive the system as an additional workload, and adapting to a new system may feel overwhelming given their existing responsibilities. To address this, it is important to clearly communicate and demonstrate that the system reduces effort by prioritising cases and standardising the workflow, making case management more efficient rather than adding complexity.

6. Learning and Reflections

One key takeaway from this project was the importance of effectively scoping the problem. Initially, the issue of school bullying felt too broad and complex to address, and our problem statement lacked clarity. Through early iterations and research, we refined our focus from broadly "reducing bullying" to a more specific and actionable challenge: helping teachers prioritise and respond to cases more effectively. This shift allowed us to design a focused solution with clear impact, rather than attempting to tackle an overly ambitious problem.

We also learned that iteration is not always linear, and that meaningful progress often requires stepping back and rethinking our approach. In the early stages, we explored broader mental health issues, drawing from general experiences such as social media. However, we realised that the scope was too large to address effectively. By narrowing our focus to specific, observable situations, we eventually arrived at bullying in schools as a more tractable problem. While it was difficult to let go of ideas, we had already invested time in, we learned that holding onto them could limit the development of more relevant solutions. Ultimately, this experience showed us that pivoting is not a setback, but a necessary step towards building a more meaningful and impactful solution.

Lastly, one of the most challenging aspects of this project was also attempting to frame a deeply human and nuanced issue within a purely data-driven approach. While many problems can be structured into patterns and predictions, bullying proved difficult to quantify due to its complexity, from the type of behaviour to how differently each child may experience it. Through our conversations with professionals from the Children's Society, we gained a deeper understanding that the impact on the child plays a crucial role in determining urgency, and that these impacts are not always immediately observable. Many of these factors are subjective and require an understanding of the child's emotional state, which cannot be fully captured through structured data alone. This highlighted the limitation of relying solely on quantifiable inputs. This experience reinforced an important insight: not everything can or should be forced into a data framework. As highlighted during our early ideation process, not all problems are purely data problems. This prompted us to reflect critically on our approach and recognise the importance of combining data-driven methods with human judgement when addressing complex social issues.

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